

CHAPTER 1

INTRODUCTION

In the modern digital era, the recruitment process has undergone a significant transformation with the advancement of online job portals and intelligent employment platforms. Every year, thousands of students, fresh graduates, and skilled professionals search for suitable jobs through various websites and applications. However, despite the availability of numerous job opportunities, candidates often face difficulty in identifying positions that truly match their personal skills, educational qualifications, technical knowledge, and career interests. As a result, job seekers frequently apply for unsuitable positions, leading to reduced efficiency, wasted time, and lower chances of successful placement.

At the same time, employers and recruiters also experience challenges in finding candidates whose profiles meet the exact skill requirements of available job roles. Traditional job searching systems mainly focus on displaying large volumes of job listings without providing personalized or intelligent recommendations. Such systems require candidates to manually search, compare, and evaluate multiple opportunities, which can be a time-consuming and confusing process, especially for freshers who are not fully aware of industry expectations.

To overcome these limitations, the present project titled “Skill + Job Matching System” has been designed and developed as an intelligent web-based application that aims to connect job seekers with relevant job opportunities based on their entered skills and profile information. The core idea behind this project is to create a smart platform that can analyze user competencies and compare them with job requirements in order to provide suitable recommendations automatically.

The system allows users to create an account, log in securely, manage their profile, upload resumes, and enter both technical as well as non-technical skills. Based on this information, the application performs matching with available job listings stored in the database and suggests the most appropriate jobs according to the similarity of required skills. This reduces manual searching effort and helps users focus only on those jobs that are most relevant to their qualifications.

The application has been developed using modern web technologies such as Python Flask Framework for backend processing, HTML, CSS, Bootstrap, and JavaScript for frontend design, and SQLite Database for efficient data storage and retrieval. The project follows a practical software development approach involving user authentication, skill input handling, resume processing, job recommendation logic, and result display.

This project is highly useful for students, fresh graduates, and beginners who often struggle to understand which jobs are suitable for them in a competitive market. By introducing intelligent skill-based matching, the proposed system acts as a bridge between available talent and employment opportunities. It not only simplifies the recruitment search process but also demonstrates the real-world implementation of database-driven web applications and recommendation-based software systems.

Thus, the Skill + Job Matching System represents an innovative and practical final year project that combines web development, user management, and intelligent job filtering into a single effective solution for modern career assistance.

CHAPTER 2

PROBLEM STATEMENT

In today's competitive employment environment, job seekers are provided with access to a large number of online recruitment websites and job portals. Although these platforms contain thousands of job vacancies, the major problem faced by users is the absence of an intelligent mechanism that can help them identify jobs according to their individual skills, educational background, and technical competencies. Most traditional systems simply display available job listings without analyzing whether a particular candidate is actually suitable for the role.

As a result, candidates are often forced to manually search through numerous job postings, read job descriptions one by one, compare required skills, and then decide whether to apply. This process is not only time-consuming but also highly inefficient. Fresh graduates and students especially face difficulty because they may not have a clear understanding of which jobs match their profile or what skills are demanded in the industry. Due to this confusion, many users end up applying for irrelevant jobs, which leads to rejection, frustration, and loss of confidence.

Another significant issue is that existing recruitment systems generally focus more on job advertisement rather than personalized career guidance. They do not provide any smart recommendation feature that can automatically filter jobs based on user-entered skills. In addition, candidates are required to repeatedly upload resumes and enter profile details on different websites, making the process repetitive and inconvenient.

From the recruiter's perspective, there is also a challenge in receiving applications from many unsuitable candidates who do not meet the required qualifications. This increases screening workload and reduces the efficiency of the recruitment process.

Therefore, there is a strong need for a web-based intelligent system that can reduce the gap between job seekers and relevant job opportunities. Such a platform should be capable of collecting candidate skill information, comparing it with job requirements, and generating suitable recommendations in a quick and user-friendly manner.

The lack of an automated skill-based recommendation platform inspired the development of this project, namely "Skill + Job Matching System," which aims to solve the problem of irrelevant job searching by introducing personalized and intelligent job matching functionality.

CHAPTER 3

PROPOSED SOLUTION

To overcome the limitations of traditional online job portals and manual job searching methods, the proposed project “Skill + Job Matching System” has been developed as an intelligent and user-friendly web application that provides personalized job recommendations according to the user’s entered skills, qualifications, and interests. The purpose of the proposed solution is to simplify the recruitment search process and help candidates identify suitable jobs in a faster and more efficient manner.

The developed system works on the concept of skill-based matching. Instead of showing all job vacancies in a general way, the application first collects important information from the user such as personal details, educational qualification, technical skills, resume information, and preferred job domain. After storing this information in the database, the system compares the user profile with the available job listings stored in the application.

Each job entry in the database contains details like job title, company name, required skills, qualification, and job description. The backend of the application processes this information and checks which jobs have maximum similarity with the user-entered skills. Based on this comparison, the system displays the most relevant job opportunities to the candidate.

The application also provides a secure registration and login module where users can create their accounts and maintain their professional profiles. After successful login, users can update their skills, upload resumes, manage profile information, and view matching job recommendations on the dashboard.

The entire project has been developed using Python for backend processing and application logic. The frontend user interface has been designed using HTML and CSS to provide a clean and simple user experience. SQLite database is used to store user records, job listings, skill details, and resume-related information. For online accessibility and deployment, the project has been hosted on the Render cloud server, which allows the web application to run efficiently through an internet browser.

Thus, the proposed solution serves as a smart digital bridge between job seekers and relevant job opportunities. It reduces unnecessary manual searching, improves the accuracy of job selection, and provides an effective demonstration of web development, database integration, and intelligent recommendation concepts.

CHAPTER 4

OBJECTIVES OF THE PROJECT

The main objective of developing the Skill + Job Matching System is to build a smart online platform that helps users find suitable job opportunities according to their skills, educational qualifications, and interests. The project is intended to reduce the complexity of traditional job searching and provide a practical solution through web technology.

The specific objectives of this project are as follows:

To develop a web-based platform for providing skill-based job recommendations.

To reduce the manual effort of searching through large numbers of irrelevant job vacancies.

To provide a secure registration and login system for personalized user access.

To store and manage candidate profiles, resumes, and job listings efficiently using SQLite database.

To compare user-entered skills with available job requirements and provide matching job suggestions.

To create an easy and user-friendly dashboard for profile management and recommendation viewing.

To implement resume upload and skill update features for better user profile analysis.

To demonstrate the practical implementation of Python programming, HTML/CSS frontend design, database connectivity, and cloud deployment.

To provide a scalable recruitment assistance system that can be further enhanced in future.

Hence, the overall objective of this project is to make the job searching process smarter, quicker, and more relevant for students, fresh graduates, and job seekers.

CHAPTER 5

LITERATURE REVIEW

The recruitment industry has experienced significant digital transformation over the past decade with the rise of online employment platforms, resume databases, and professional networking websites. Traditional job searching methods such as newspaper advertisements, manual applications, and offline consultancy services have gradually been replaced by web-based job portals that provide quick access to thousands of vacancies. However, despite this technological growth, many existing systems still lack personalized recommendation capabilities and intelligent skill-based filtering.

Several online job portals available today mainly function as vacancy advertisement platforms where employers publish job openings and candidates manually search according to category, location, or company name. Although these systems provide large job databases, they do not fully address the issue of matching candidate skills with recruiter requirements. As a result, users are often required to browse through many irrelevant listings before finding suitable opportunities.

Research in the field of recommendation systems shows that personalized matching platforms improve user efficiency by analyzing profile data and suggesting relevant content automatically. Similar recommendation approaches are widely used in e-commerce, digital media, and educational platforms. Applying the same concept to recruitment can significantly improve job search quality by connecting candidate competencies with job demands.

Existing studies also indicate that fresh graduates and students face the highest level of confusion during job searching because they often possess limited industry knowledge and are unable to identify which positions are best suited for their skill set. This creates a need for an intelligent guidance system that can provide more focused employment suggestions rather than displaying all vacancies in a generalized form.

Many modern career platforms have started integrating resume parsing, profile scoring, and recommendation features, but such systems are often complex, expensive, or dependent on large-scale Artificial Intelligence infrastructure. For academic and educational implementation, there is still a need for a lightweight, simple, and effective system that can demonstrate the concept of skill-based recruitment assistance using basic web development technologies and database management.

The present project “Skill + Job Matching System” has been developed after studying these limitations and opportunities in existing job search systems. It attempts to provide a simplified intelligent recommendation mechanism where user-entered skills are compared with job requirements stored in the database to generate relevant results. This makes the system more personalized, efficient, and beneficial for students and fresh job seekers.

Thus, the literature survey clearly indicates that intelligent recommendation-based recruitment systems are becoming increasingly important, and the proposed project is a practical academic implementation of this modern concept.

CHAPTER 6

TOOLS AND TECHNOLOGIES USED

The successful development of the Skill + Job Matching System required the integration of various software tools, programming technologies, database systems, and hosting services. Each technology used in this project plays a specific role in ensuring smooth frontend design, backend processing, data management, and online deployment. The major tools and technologies used are described below:

6.1 Python

Python is the main programming language used for backend development and application logic in this project. It is a high-level, interpreted, and user-friendly programming language widely used for web development, data handling, automation, and software applications.

In the present system, Python is used for:

- Handling user registration and login authentication
- Processing user profile information
- Managing resume upload functionality
- Comparing user-entered skills with job requirements
- Generating matching job recommendations
- Connecting the web pages with the SQLite database

Python provides simplicity, flexibility, and faster development, which makes it highly suitable for educational and real-time web projects.

6.2 HTML

HTML (Hyper Text Markup Language) is used to create the structure and content of all web pages in the application. It defines the layout of forms, buttons, text fields, tables, headings, and dashboard sections visible to the user.

HTML pages are used in this project for:

- Login page
- Registration page
- User dashboard
- Profile page
- Resume upload page
- Job recommendation result page

Thus, HTML acts as the basic skeleton of the user interface.

6.3 CSS

CSS (Cascading Style Sheets) is used to design and beautify the web pages created with HTML. It improves the visual appearance, spacing, alignment, colors, fonts, and responsiveness of the application.

In this project, CSS is used to provide:

- Attractive page styling
- Better button and form design
- Layout alignment
- User-friendly dashboard interface
- Consistent look across all pages

CSS helps in making the application more professional and easier to use.

6.4 SQLite Database

SQLite is used as the database management system for storing and retrieving all project-related information. It is a lightweight and serverless relational database that is highly suitable for small to medium web applications.

The SQLite database stores:

- User registration details
- Login credentials
- Resume information
- User skills
- Job vacancy details
- Matching records

SQLite provides easy integration with Python and ensures fast access to stored records without requiring a separate database server.

6.5 Render Cloud Hosting

Render is used as the online cloud hosting platform for deploying the web application on the internet. After development on the local system, the project is uploaded and hosted on Render so that it can be accessed through a web browser from anywhere.

Render provides:

- Online server deployment
- Continuous application availability

- Easy GitHub integration
- Automatic project updates
- Secure cloud execution

Using Render hosting makes the project function like a real-world online web application rather than only a local software program.

6.6 GitHub Repository

GitHub is used for storing the project source code online and maintaining version control. It helps in saving all files including Python code, HTML templates, CSS files, and database configuration in a centralized repository.

GitHub is useful for:

- Code backup
- Version management
- Easy deployment with Render
- Future modifications and updates

Thus, the combination of Python, HTML, CSS, SQLite, Render, and GitHub has enabled the successful development and deployment of an efficient skill-based online job recommendation system.

CHAPTER 7

SYSTEM REQUIREMENTS

For the successful development, testing, and deployment of the Skill + Job Matching System, certain hardware and software resources were required. These system requirements ensure smooth execution of the application, proper database handling, and uninterrupted user interaction. The requirements are categorized as follows:

7.1 Hardware Requirements

The hardware requirements used for developing this project are simple and easily available on a normal personal computer or laptop. Since the project is a lightweight web-based application, it does not require any high-end system configuration.

The minimum hardware requirements are:

- Processor: Intel Core i3 or higher
- RAM: 4 GB minimum
- Hard Disk Space: 500 MB free space
- Keyboard and Mouse
- Display Monitor
- Internet Connection for deployment and online access

A system with the above configuration is sufficient for coding, database handling, testing, and running the application.

7.2 Software Requirements

The software tools used during the development and deployment of this project are as follows:

Sr. No.	Software / Tool	Purpose
1.	Python	Backend programming and application logic
2.	HTML	Web page structure design
3.	CSS	Web page styling
4.	SQLite	Database management
5.	VS Code / Any Code Editor	Writing and editing source code
6.	GitHub	Source code storage and version control
7.	Render	Online deployment server
8.	Web Browser	Running and testing the application

7.3 Browser Requirements

As the project is a web-based application, it can be accessed using any modern internet browser. The recommended browsers are:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

These browsers support HTML, CSS rendering, form handling, and cloud-hosted web application access smoothly.

7.4 Internet Requirements

An active internet connection is necessary for:

- Uploading the project on GitHub
- Deploying on Render server
- Accessing the live web application online
- Downloading dependencies and updates

Without internet connectivity, only the local offline version can be tested.

7.5 User Requirements

To use the Skill + Job Matching System efficiently, the user must have:

- Basic knowledge of internet browsing
- Ability to create account/login
- Understanding of personal skills and resume details
- Basic familiarity with online job applications

This makes the system suitable for students, freshers, and general job seekers.

Thus, the above hardware and software requirements are sufficient for the successful implementation and execution of the proposed project.

CHAPTER 8

EXISTING SYSTEM ANALYSIS

Before the development of the proposed Skill + Job Matching System, it was necessary to study the existing methods and online platforms currently used by candidates for searching jobs. The analysis of the present system helps in understanding the drawbacks, limitations, and practical problems faced by users in traditional job searching environments.

At present, most candidates depend on various online job portals, company career pages, social media advertisements, and employment websites for finding job opportunities. These platforms usually provide a large number of vacancies categorized according to job title, company, location, and salary

In the existing system, candidates have to manually perform the following tasks:

- Search job listings one by one
- Read every job description carefully
- Compare required qualifications with personal skills
- Decide manually whether the job is suitable or not
- Repeat the same process for multiple vacancies

This makes the job searching process lengthy, repetitive, and inefficient. Fresh graduates often face more difficulty because they do not have enough practical knowledge regarding industry skill demands.

Another major drawback of existing systems is that they generally display jobs in a broad category format rather than offering skill-oriented recommendations. A user searching for employment may still receive many unrelated jobs, which creates confusion and wastes time.

Existing job portals also require repeated profile filling and resume uploads on different websites, which can become frustrating for users. Moreover, many platforms are advertisement-focused and not guidance-focused, meaning they are designed more for vacancy display than for intelligent career assistance.

Therefore, after analyzing the existing systems, the following major limitations were identified:

- Limitations of Existing System
- Lack of personalized skill-based job recommendation
- Time-consuming manual job searching
- Repetitive profile submission process
- Difficulty for freshers in identifying suitable jobs
- Presence of many irrelevant vacancies
- No direct intelligent matching between candidate and job requirement

These limitations clearly show the need for a more focused and efficient platform that can automatically compare user skills with available vacancies and provide relevant recommendations. This requirement led to the development of the proposed Skill + Job Matching System.

CHAPTER 9

PROPOSED SYSTEM DESIGN

The Skill + Job Matching System has been designed as an intelligent web-based recruitment assistance platform that focuses on providing relevant job opportunities according to the user's entered skills and profile details. The proposed system is designed to overcome the drawbacks of the traditional job searching process by introducing automated matching, user profile management, and a centralized recommendation dashboard.

The overall architecture of the system follows a simple client-server model where the user interacts with the web pages through a browser, the backend processes user requests using Python, and the SQLite database stores all important records such as candidate profiles, skills, resumes, and job listings.

The proposed system consists of the following major design components:

9.1 User Interface Design

The frontend of the system is developed using HTML and CSS. It provides a clean and user-friendly interface through which candidates can easily interact with the application.

The main pages designed in the system include:

- Home Page
- User Registration Page
- User Login Page
- Dashboard
- Profile Update Page
- Resume Upload Page
- Job Recommendation Page
- Logout Module

The design is kept simple so that even a beginner user can access the application without technical difficulty.

9.2 Backend Processing Design

The backend processing of the project is implemented in Python. It handles all major functionalities such as:

- User authentication
- Database connection
- Resume information handling
- Skill storage
- Job comparison logic

- Recommendation result generation

Whenever the user enters or updates skills, the backend compares this information with the required skills present in the job database and displays the most relevant matches.

9.3 Database Design

SQLite database is used for efficient storage and retrieval of records. The database contains multiple tables for:

- User Details
- Login Credentials
- Skills Information
- Resume Records
- Job Listings
- Match Results

This database structure allows the application to access user and job information quickly whenever recommendations are generated.

9.4 Skill Matching Mechanism

The most important part of the proposed design is the skill matching mechanism. In this process:

- User enters technical skills and qualifications.
- Job database contains required skills for each vacancy.
- Python backend compares both records.
- Similarity-based matching jobs are selected.
- Recommended jobs are displayed on user dashboard.

This makes the system more efficient than general vacancy display systems.

9.5 Online Deployment Design

To make the project accessible through the internet, the complete application is deployed on the Render cloud server. This enables users to access the platform online using a browser without depending only on local machine execution.

GitHub repository is connected with Render for code hosting and deployment updates.

9.6 Advantages of Proposed Design

The proposed system design offers the following benefits:

- Centralized user profile management
- Faster job searching process
- Personalized recommendation
- Easy database maintenance
- Cloud accessibility
- User-friendly interface

Thus, the proposed system design is practical, organized, and efficient for implementing a real-time skill-based online job recommendation platform.

CHAPTER 10

MODULES OF SKILL + JOB MATCHING SYSTEM

The Skill + Job Matching System is divided into several functional modules to ensure smooth working, proper user interaction, secure data handling, and efficient job recommendation. Each module performs a specific task and together they form the complete workflow of the application. The major modules of the system are described below:

10.1 User Registration Module

This module allows new users to create an account on the platform by entering basic personal details such as:

- Full Name
- Email ID
- Username
- Password
- Qualification

The registration data is stored in the SQLite database. This module ensures that every candidate has a unique account through which personalized job recommendations can be provided.

Functions of Registration Module:

- Collects new user information
- Stores user data securely
- Prevents duplicate account creation
- Enables personalized access

10.2 User Login Module

After successful registration, users can access the application using their username/email and password. The login module verifies entered credentials with the database and grants secure access to the dashboard.

Functions of Login Module:

- Authenticates registered users
- Provides secure access
- Prevents unauthorized login
- Redirects users to dashboard

This module is essential for maintaining account privacy and user-specific records.

10.3 User Profile Management Module

This module allows users to manage and update their profile information after login. Users can enter or edit:

- Personal details
- Educational qualification
- Technical skills
- Preferred job category
- Contact information

Updated information helps the system provide more accurate job recommendations.

Functions:

- Profile editing
- Skill updating
- Qualification management
- User information maintenance

10.4 Resume Upload Module

The resume upload module allows users to upload their resume or professional document for maintaining complete profile information within the system.

This module helps in:

- Storing resume files
- Maintaining professional records
- Improving profile completeness
- Future enhancement for resume analysis

Uploaded resume details are linked with the respective user account.

10.5 Job Database Management Module

This module stores all available job listings inside the SQLite database. Each job entry contains:

- Job Title
- Company Name
- Required Skills
- Qualification
- Job Description

This serves as the core job source from which recommendations are generated.

Functions:

- Add job vacancies
- Store job requirements
- Maintain organized job records
- Retrieve jobs during matching
- 10.6 Skill Matching Module

This is the most important intelligent module of the entire project. In this module, the system compares the user-entered skills with the required skills present in job listings.

The Python backend checks:

- matching keywords
- qualification suitability
- profile relevance

and then selects jobs that have the highest compatibility.

Functions:

- Skill comparison
- Similarity checking
- Relevant job filtering
- Recommendation generation

This module is responsible for the smart behavior of the application.

10.7 Recommendation Display Module

After the matching process is completed, all relevant job opportunities are displayed on the user dashboard in a structured form.

Displayed information includes:

- Job Title
- Company Name
- Required Skills
- Qualification

- Job Description

This module helps users quickly view the jobs that are best suited for them.

10.8 Cloud Deployment Module

The complete web application is hosted on the Render cloud platform, which makes the project accessible online.

This module ensures:

- online availability
- internet access
- remote testing
- deployment management

GitHub integration helps in updating the source code whenever changes are made.

Thus, the combination of all these modules makes the Skill + Job Matching System a complete and functional recruitment assistance web application.

CHAPTER 11

DATABASE DESIGN

Database design is one of the most important components of the **Skill + Job Matching System** because the entire application depends upon storing, retrieving, updating, and comparing user information with available job listings. In this project, **SQLite** has been used as the database management system due to its lightweight structure, easy integration with Python, and efficient record handling.

The database is designed in such a way that it can manage all essential information related to users, resumes, job vacancies, and matching details. Multiple tables are created in SQLite for systematic data storage.

The major database tables used in this project are described below:

11.1 User Registration Table

This table stores the registration details of every candidate who creates an account on the platform.

Field Name	Data Type	Description
------------	-----------	-------------

user_id	Integer	Unique ID of user
full_name	Text	Name of candidate
email	Text	Email address
username	Text	Login username
password	Text	User password
qualification	Text	Educational qualification

This table helps in identifying each registered user separately.

11.2 User Profile / Skills Table

After login, users can update their profile and add skills. This information is stored in the profile table.

Field Name	Data Type	Description
------------	-----------	-------------

profile_id	Integer	Unique profile ID
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Field Name	Data Type	Description
user_id	Integer	Linked user ID
technical_skills	Text	Candidate technical skills
non_technical_skills	Text	Other skills
preferred_domain	Text	Preferred job field
contact_info	Text	Contact details

This table is used during skill matching.

11.3 Resume Upload Table

This table stores the uploaded resume information of users.

Field Name	Data Type	Description
resume_id	Integer	Unique resume ID
user_id	Integer	Linked user account
resume_name	Text	Resume file name
resume_path	Text	Resume storage location

This module keeps professional documents linked with the candidate profile.

11.4 Job Listings Table

This is the most important data table containing all available jobs.

Field Name	Data Type	Description
job_id	Integer	Unique job ID
job_title	Text	Name of job
company_name	Text	Hiring company
required_skills	Text	Skills required

Field Name	Data Type	Description
qualification_required	Text	Qualification needed
job_description	Text	Details of vacancy

This table acts as the source for recommendation generation.

11.5 Match Result Table

This table can be used to store matching history or recommended jobs shown to users.

Field Name	Data Type	Description
match_id	Integer	Unique match ID
user_id	Integer	Candidate ID
job_id	Integer	Recommended job ID
match_score	Text	Matching percentage/result

This improves recommendation tracking.

11.6 Database Relationship

The overall database relationship works as follows:

- One user account is linked with one profile.
- One user can upload one or more resumes.
- Job listings are stored separately.
- User skills are compared with job listings.
- Matching results are generated based on database comparison.

Thus, the SQLite database creates a centralized record management system for the complete application.

11.7 Advantages of SQLite in this Project

SQLite was selected because:

- It is lightweight and fast
- No separate server installation required
- Easy integration with Python
- Suitable for academic web projects

- Supports structured table creation
- Easy backup and maintenance

Hence, the database design provides strong backend support for the successful functioning of the Skill + Job Matching System.

CHAPTER 12

PROJECT FOLDER STRUCTURE

The Skill + Job Matching System is organized into multiple folders and files so that the source code remains clean, maintainable, and easy to understand. Proper folder structuring is an essential part of software development because it separates backend logic, frontend pages, styling files, database files, and uploaded documents in a systematic manner.

In this project, all source files are arranged according to their functionality. The folder structure helps the developer in managing code efficiently, performing updates, fixing bugs, and deploying the project on the Render server.

A generalized structure of the project is shown below:

skill-job-matcher/

```
|
|
|— app.py
|— requirements.txt
|— render.yaml
|— jobs.db
|
|— static/
|   |— style.css
|   |— dashboard.css
|   |— images/
|
|— templates/
|   |— index.html
|   |— register.html
|   |— login.html
|   |— dashboard.html
```

```
| |— profile.html
| |— upload_resume.html
| |— recommendations.html
| |— result.html
|
|— uploads/
|   |— user_resumes/
|
|— README.md
```

12.1 app.py

This is the main Python application file that contains the complete backend logic of the system. It manages:

- route handling
- user registration
- login authentication
- database connection
- skill matching logic
- recommendation generation

The entire application runs through this core file.

12.2 requirements.txt

This file contains the list of Python dependencies and libraries required to run the project on local machine as well as on Render deployment server.

It ensures that all necessary packages are installed automatically during hosting.

12.3 render.yaml

This file is used for deployment configuration on the Render cloud platform. It defines the build commands, start commands, and server settings required for successful online hosting of the project.

12.4 jobs.db (SQLite Database File)

This is the SQLite database file in which all important records are stored such as:

- user details
- login credentials
- skills
- uploaded resumes
- job listings
- recommendation data

This file acts as the data storage center of the entire application.

12.5 static Folder

The **static** folder contains all CSS design files and images used in the frontend.

Main contents include:

- styling sheets
- dashboard designs
- icons/images
- visual assets

This folder is responsible for the visual appearance of the website.

12.6 templates Folder

The **templates** folder contains all HTML pages that are displayed to the user through browser interaction.

These pages include:

- home page
- registration form
- login page
- dashboard
- profile update page
- resume upload page
- recommendation results

Each user request is routed to these templates by the backend Python application.

12.7 uploads Folder

This folder is used for storing the resumes uploaded by candidates. It acts as the document storage area of the project.

Uploaded files are linked with user accounts and can be processed whenever required.

12.8 README.md

This file contains the project description, setup instructions, GitHub information, and developer notes.

It helps in understanding the project workflow and deployment process.

12.9 Advantages of Organized Folder Structure

A proper folder structure provides the following benefits:

- Easy code maintenance
- Better readability
- Faster debugging
- Simplified deployment
- Professional project management
- Future scalability

Hence, the project folder structure plays an important role in keeping the Skill + Job Matching System technically organized and development-friendly.

CHAPTER 13

WORKING METHODOLOGY / SYSTEM FLOW

The **Skill + Job Matching System** works through a systematic sequence of user interaction, backend processing, database comparison, and recommendation generation. The complete methodology is designed to ensure that a user can easily create a profile, enter skill details, and receive suitable job suggestions without manually browsing through hundreds of unrelated vacancies.

The working flow of the proposed system is explained below step by step:

13.1 User Registration

The first step in the working methodology is **user registration**. A new candidate visits the web application and creates an account by entering required personal details such as:

- Name
- Email ID
- Username
- Password
- Qualification

These details are validated and stored securely in the SQLite database.

This step ensures that every user has an individual account for personalized recommendation access.

13.2 User Login Authentication

After successful registration, the candidate can log in using the registered username and password.

The Python backend verifies:

- whether the account exists,
- whether password is correct,
- and whether the login request is authentic.

After successful verification, the user is redirected to the personal dashboard.

This module ensures security and private profile handling.

13.3 Profile Completion and Skill Entry

Once logged in, the user is required to complete profile information by entering:

- educational details,
- technical skills,
- preferred job domain,
- contact details,
- additional profile information.

The user can also upload resume documents for profile completeness.

This information is then saved into the database and becomes the primary source for job comparison.

13.4 Job Vacancy Storage in Database

On the backend side, the system already contains multiple job listings stored in SQLite database. Each vacancy contains:

- job title,
- company name,
- required skills,
- qualification requirement,
- job description.

This database acts as the reference source against which user skills are checked.

13.5 Skill Matching Process

This is the core functional step of the project.

After the user enters or updates skills, the Python backend performs a comparison between:

candidate skill profile

and

required job skills stored in database.

The system identifies:

- exact skill matches,
- similar qualification matches,
- preferred domain relevance.

Based on this comparison, only those jobs that satisfy the user profile are selected.

This recommendation workflow follows the same practical idea used in modern skill-based job recommendation platforms, where user competencies are matched against vacancy requirements to improve relevance and reduce manual searching.

13.6 Recommendation Generation

After successful comparison, the filtered and matched job opportunities are generated by the backend.

The user receives a list of suitable jobs containing:

- Job Title
- Company Name
- Required Skills
- Qualification
- Description

Thus, instead of manually checking every vacancy, the user directly gets relevant employment suggestions.

13.7 Dashboard Display

All recommended jobs are displayed on the user dashboard in an organized and readable form.

The dashboard also allows:

- profile update,
- skill modification,
- resume management,
- repeated recommendation checking.

This creates an interactive user experience.

13.8 Online Access through Render Server

Since the complete application is deployed on Render cloud hosting, the user can access the platform online through a browser from any internet-connected device.

This makes the project behave like a real-world live web application rather than only a local machine project.

13.9 Overall System Flow Summary

The complete working flow can be summarized as:

Registration → Login → Profile Completion → Skill Entry → Database Comparison → Job Matching → Recommendation Display

This structured methodology ensures that the application remains simple for users while performing intelligent filtering in the backend.

CHAPTER 14

KEY FEATURES OF THE APPLICATION

The **Skill + Job Matching System** has been developed with several practical features that make the application useful, user-friendly, and efficient for candidates searching for relevant employment opportunities. These features are designed to simplify profile management, improve recommendation accuracy, and provide smooth online access to users.

The major key features of the application are explained below:

14.1 Secure User Registration and Login

The application provides a secure registration and login mechanism through which every user can create a personalized account.

This feature ensures:

- unique account creation,
- user authentication,
- private profile storage,
- secure access to recommendations.

Only registered users can access the dashboard and use recommendation services.

14.2 Personalized User Dashboard

After successful login, each user is provided with a dedicated dashboard where all major options are available in one place.

The dashboard allows users to:

- view profile,
- update skills,
- upload resume,
- check recommended jobs,
- manage account details.

This improves convenience and user interaction.

14.3 Skill-Based Job Recommendation

This is the most important feature of the project.

Instead of displaying all job vacancies, the system analyzes user-entered technical skills and compares them with job requirements available in the database.

Only the most relevant jobs are recommended.

This feature helps users:

- save time,
- avoid irrelevant vacancies,
- focus on suitable opportunities.

14.4 Resume Upload Facility

Users can upload their resume within the application, which helps in maintaining complete professional profile information.

This feature provides:

- document storage,
- profile completeness,
- better future enhancement possibilities.

14.5 Profile Management Feature

The system allows candidates to update their:

- qualification,
- skills,
- contact details,
- preferred job field.

This ensures that job recommendations remain accurate according to latest user information.

14.6 Database-Driven Job Storage

All available job opportunities are stored in the SQLite database in a structured format.

This allows:

- quick job retrieval,
- easy vacancy management,
- efficient comparison with candidate profile.

14.7 Online Cloud Deployment

The project is deployed on Render cloud hosting, making it accessible online through internet browsers.

This feature gives the application:

- real-time online availability,
- live testing capability,
- remote user access.

It also demonstrates practical cloud deployment knowledge.

14.8 Simple and User-Friendly Interface

The frontend of the application is developed using HTML and CSS with a simple layout that can be easily used by students, freshers, and non-technical users.

The interface provides:

- easy navigation,
- clean forms,
- readable output,
- smooth user interaction.

14.9 Efficient Data Handling

Since SQLite is used as backend storage, the application can efficiently maintain:

- user records,
- skill data,
- resumes,
- job listings,
- recommendation information.

This ensures smooth backend performance.

14.10 Scalable Future Expansion

The current project structure is flexible and can be upgraded in future with:

- AI-based resume analysis,
- machine learning recommendations,

- company recruiter panel,
- live API-based jobs.

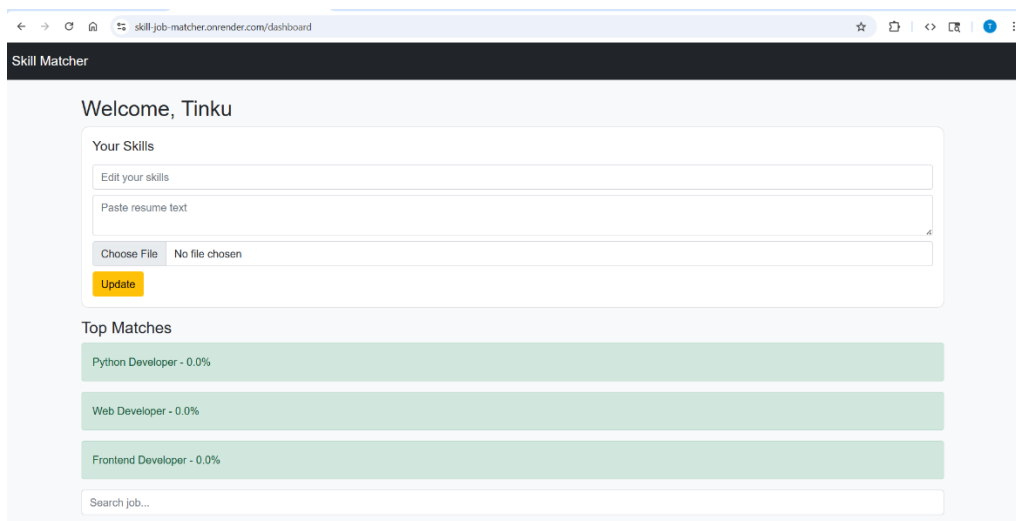
Thus, the application has strong future enhancement potential.

CHAPTER 15

OUTPUT SCREENS / RESULT ANALYSIS

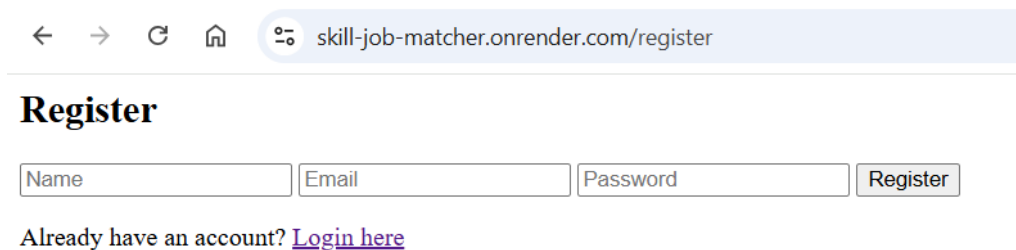
The successful implementation of the **Skill + Job Matching System** can be verified through the following output screens. Each screen represents a major module of the application and demonstrates the practical working of user registration, login, profile handling, skill matching, and recommendation generation.

15.1 Home Page Output



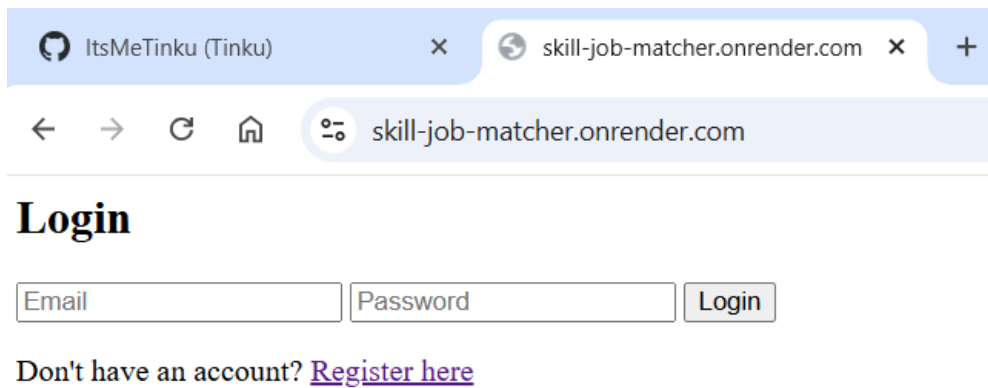
The home page is the introductory interface of the application through which users can understand the purpose of the platform and navigate to registration or login options. It provides the first interaction point between the user and the system. The design is kept simple and attractive for easy accessibility.

15.2 User Registration Page Output



The registration page allows new candidates to create their account by entering required personal information such as name, email, username, password, and qualification. After successful submission, the data is stored in the SQLite database for future authentication and personalized access.

15.3 User Login Page Output



ItsMeTinku (Tinku) skill-job-matcher.onrender.com

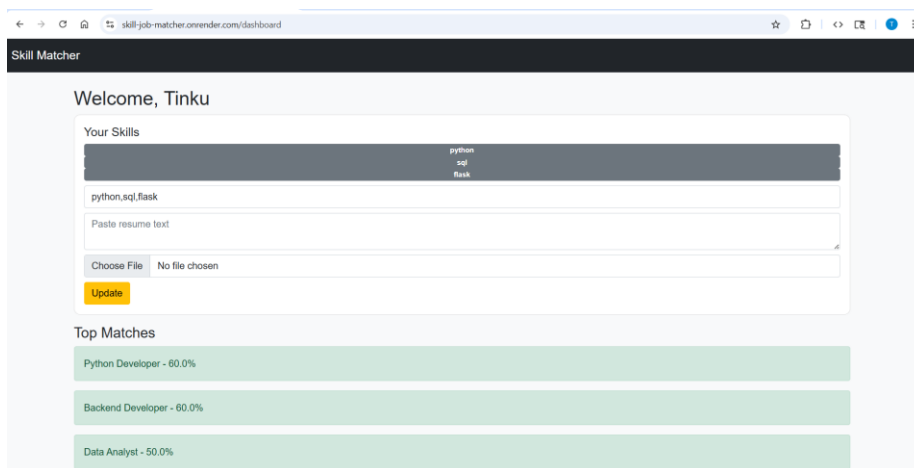
← → ↻ 🏠 🔍 skill-job-matcher.onrender.com

Login

Don't have an account? [Register here](#)

The login page provides secure access to already registered users. The entered username and password are verified by the backend Python code. Only valid users are redirected to the dashboard, ensuring account security and user privacy.

15.4 User Dashboard Output



skill-job-matcher.onrender.com/dashboard

Skill Matcher

Welcome, Tinku

Your Skills

python
sql
flask

python,sql,flask

Paste resume text

Choose File No file chosen

Update

Top Matches

Python Developer - 60.0%

Backend Developer - 60.0%

Data Analyst - 50.0%

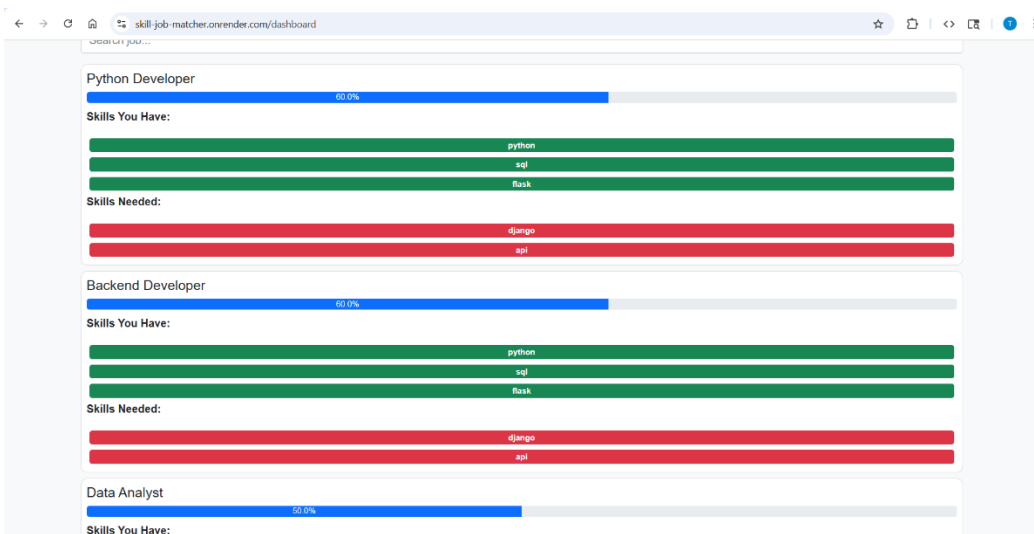
After login, the dashboard is displayed as the main control panel of the user. It contains all important modules such as profile management, skill updating, resume upload, and job recommendation viewing. This page acts as the central working area of the application.

15.5 Profile Update / Skill Entry Output



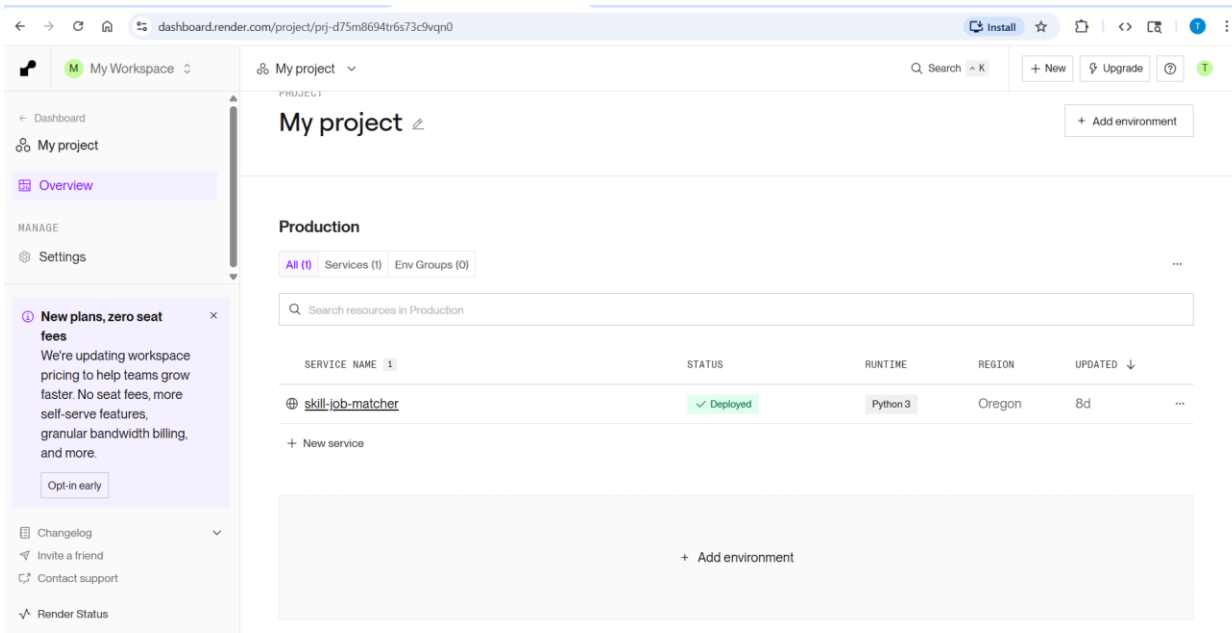
This output screen shows the module where the user enters educational details, technical skills, preferred domain, and other profile information. These details are essential for generating relevant job matches and improving recommendation accuracy.

15.6 Job Recommendation Result Output



This is the most important output screen of the project. After comparing user skills with job requirements stored in the database, the system generates a list of matching jobs. The user can view relevant vacancies along with required skills, qualification, and company details.

15.7 Online Hosted Project Output



This output verifies that the project has been successfully deployed on the Render cloud hosting platform and can be accessed online through an internet browser. This demonstrates the real-world implementation and cloud accessibility of the application.

15.8 Overall Result Analysis

The output screens clearly show that all modules of the Skill + Job Matching System are functioning successfully. The application is able to:

- register users,
- authenticate login,
- store profile data,
- upload resumes,
- compare skills,
- generate matching job opportunities,
- and provide live online access.

Thus, the final output confirms that the proposed project has been implemented effectively and achieves its objective of providing intelligent skill-based job recommendations.

CHAPTER 16

ADVANTAGES OF THE PROPOSED SYSTEM

The **Skill + Job Matching System** offers several advantages over traditional manual job searching methods and general online vacancy browsing platforms. The proposed application is designed to provide a more focused, efficient, and user-friendly recruitment assistance experience. By integrating user profile management with intelligent job recommendation, the system becomes highly beneficial for students, fresh graduates, and general job seekers.

The major advantages of the proposed system are as follows:

16.1 Reduces Manual Job Searching Effort

In traditional job portals, candidates are required to search through a large number of vacancies manually. This consumes time and often leads to confusion. The proposed system automatically filters relevant jobs according to the user's skills, reducing unnecessary browsing.

16.2 Provides Skill-Based Personalized Recommendations

One of the biggest advantages of the application is that it does not display all jobs randomly. Instead, it provides only those job opportunities that match the user's entered technical skills, qualification, and profile information.

This improves recommendation relevance and saves user effort.

16.3 User-Friendly Interface

The frontend design of the system is simple and easy to use. Even a beginner or non-technical user can easily:

- register,
- log in,
- update profile,
- upload resume,
- check recommendations.

The application does not require advanced technical knowledge.

16.4 Secure User Account Handling

Through user registration and login authentication, every candidate gets a private and secure profile space. This ensures that personal information, resumes, and recommendation data remain linked with the correct user account.

16.5 Centralized Profile Management

Users do not need to repeatedly enter the same information on multiple websites. The system stores:

- profile details,
- educational qualifications,
- technical skills,
- resumes

in one centralized platform.

16.6 Faster Job Identification

Since the backend compares user profile with job listings automatically, the user can quickly identify suitable opportunities without reading each job description manually.

This improves efficiency and reduces job search frustration.

16.7 Database-Driven Accuracy

SQLite database provides organized storage of both user details and job vacancies. This ensures:

- faster retrieval,
- structured data management,
- smoother recommendation process.

16.8 Online Accessibility

As the project is deployed on Render cloud hosting, it can be accessed online through an internet browser from any internet-enabled device. This makes the project practically usable like a real-world web application.

16.9 Easy Maintenance and Scalability

The project uses a simple but organized folder structure, which makes future modification easy. New jobs, new features, and advanced recommendation logic can be added without rebuilding the complete system.

16.10 Educational and Practical Value

Apart from user benefits, this project also demonstrates practical implementation of:

- Python backend programming,
- HTML/CSS web design,
- SQLite database connectivity,
- cloud deployment.

Hence, it serves as a strong academic and technical learning model.

Thus, the proposed system provides a smarter, faster, and more practical solution for recruitment assistance compared to existing manual job searching approaches.

CHAPTER 17

LIMITATIONS OF THE SYSTEM

Although the **Skill + Job Matching System** has been successfully developed and performs its intended functions effectively, there are still certain limitations in the current version of the application. These limitations are mainly due to the academic scope, simplified implementation, and time constraints of the final year project. Identifying these limitations is important because it helps in understanding the boundaries of the present system and also provides a direction for future enhancement.

The major limitations of the current system are as follows:

17.1 Limited Job Database

The recommendations generated by the application depend upon the job listings manually stored in the SQLite database. Since the current project uses a limited academic dataset, the number of available jobs is not as large as commercial job portals.

Therefore, recommendation diversity is limited to the jobs present in the database.

17.2 Basic Skill Matching Logic

The current recommendation process works on direct comparison between user-entered skills and job-required skills. Although effective, it does not use advanced Artificial Intelligence or Machine Learning algorithms for deeper semantic understanding.

As a result, recommendation accuracy is based on keyword similarity rather than intelligent predictive analytics.

17.3 No Recruiter/Admin Panel

At present, the project mainly focuses on candidate-side interaction. There is no fully developed recruiter or admin panel through which companies can dynamically add vacancies, manage applicants, or perform direct recruitment actions.

Job entries are managed at the backend level.

17.4 Limited Resume Analysis

The system allows users to upload resumes, but the uploaded document is mainly stored as profile data. The current version does not perform advanced automatic resume parsing or detailed resume scoring.

17.5 Internet Dependency for Hosted Version

Since the live application is deployed on Render cloud hosting, internet access is necessary for online use. Without internet connectivity, only the local offline version can be executed.

17.6 Basic Security Level

The project includes login authentication and secure user separation, but it does not yet contain advanced enterprise-level security features such as:

- OTP verification,
- encrypted password hashing,
- multi-layer authentication.

These can be added in future versions.

17.7 Academic Scale Implementation

Being a final year academic project, the system is designed as a functional prototype rather than a full commercial recruitment product. Therefore, some real-world integrations such as live company APIs, email alerts, and automated application submission are not included.

Despite these limitations, the current project successfully demonstrates the concept of intelligent skill-based job recommendation and provides a strong foundation for future expansion into a more advanced recruitment platform.

CHAPTER 18

FUTURE SCOPE

The current version of the **Skill + Job Matching System** has been developed as a functional and efficient academic web application that demonstrates the concept of skill-based job recommendation. However, with the continuous growth of online recruitment technologies and intelligent software systems, there are several opportunities through which this project can be further improved and expanded in the future.

The major future enhancement possibilities are discussed below:

18.1 Integration of Artificial Intelligence and Machine Learning

In the present system, job recommendations are generated through direct skill comparison. In future, advanced **Artificial Intelligence (AI)** and **Machine Learning (ML)** algorithms can be integrated to provide smarter and more accurate recommendations.

AI can help in:

- predictive job suitability analysis,
- intelligent profile scoring,
- semantic skill understanding,
- personalized career suggestions.

This will improve recommendation quality significantly.

18.2 Automatic Resume Parsing

Currently, resumes are uploaded mainly for record maintenance. In future versions, automatic resume parsing can be introduced so that the system can extract:

- education,
- skills,
- certifications,
- experience

directly from uploaded resume files.

This will reduce manual data entry by the user.

18.3 Recruiter / Company Panel

A separate recruiter or company-side panel can be developed where employers can:

- create accounts,
- upload vacancies,
- search suitable candidates,
- review matching applicants.

This will make the project a two-way recruitment platform instead of only candidate-side recommendation software.

18.4 Live Job API Integration

At present, job listings are stored manually in SQLite database. In future, the project can be connected with live online job APIs so that latest vacancies from real recruitment websites can be fetched automatically.

This will increase:

- job variety,
- recommendation freshness,
- practical usefulness.

18.5 Email and Notification Alerts

Future versions can include automated notification systems such as:

- email alerts,
- recommended job reminders,
- profile update notifications.

This will improve user engagement and keep candidates informed about new opportunities.

18.6 Enhanced Security Features

More advanced security measures can be added such as:

- password encryption,
- OTP verification,
- secure file handling,
- account recovery system.

This will make the application more secure for large-scale deployment.

18.7 Advanced Dashboard Analytics

A more intelligent dashboard can be introduced showing:

- profile completion score,
- recommendation percentage,
- missing skills analysis,
- top demanded technologies.

This will provide users with deeper career insights.

18.8 Mobile Responsive / Android Version

The system can also be converted into:

- mobile responsive progressive web app, or
- Android application

so that users can access recommendations more conveniently through smartphones.

Thus, the proposed Skill + Job Matching System has strong future scalability and can be transformed into a more advanced, intelligent, and commercially usable online recruitment assistance platform.

CHAPTER 19

CONCLUSION

The **Skill + Job Matching System** has been successfully designed, developed, and implemented as a web-based final year project with the objective of simplifying the job searching process for candidates by providing skill-oriented employment recommendations. The project addresses one of the most common problems faced by students, fresh graduates, and job seekers, namely the difficulty of identifying suitable jobs among a large number of available vacancies.

Through this application, a user can create a personal account, securely log in, update profile information, upload resume details, enter technical skills, and receive matching job opportunities based on the data stored in the system. By comparing candidate skills with job requirements present in the SQLite database, the backend Python logic generates filtered recommendations that are more relevant than ordinary manual vacancy browsing.

The project has successfully demonstrated the practical integration of:

- Python programming,
- HTML and CSS frontend development,
- SQLite database management,
- cloud deployment using Render.

All major modules such as registration, login authentication, profile management, resume upload, skill comparison, and recommendation display have been implemented effectively. The online deployment of the application further enhances its practical usability and shows that the project is not limited to local execution only.

This final year project has provided valuable understanding of real-time web development, database connectivity, online hosting, and recommendation-based software logic. It also highlights how modern recruitment assistance systems can be made more efficient by introducing personalized filtering instead of generalized job listing methods.

Although the present version has certain academic limitations, it successfully fulfills its main objective of creating a functional and intelligent skill-based job recommendation platform. The project serves as a strong foundation for future enhancements involving Artificial Intelligence, live recruitment integration, and advanced analytics.

Hence, it can be concluded that the **Skill + Job Matching System** is a useful, practical, and innovative web application that provides an effective digital solution for smarter and more relevant job searching.

CHAPTER 20

BIBLIOGRAPHY / REFERENCES

The following books, online resources, websites, and technical documentations were referred during the analysis, development, coding, testing, and deployment of the Skill + Job Matching System project:

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- HTML and CSS Web Design Tutorials – Available through W3Schools: <https://www.w3schools.com/>
- SQLite Database Documentation – Available at: <https://www.sqlite.org/docs.html>
- Render Cloud Hosting Documentation – Available at: <https://render.com/docs>
- GitHub Version Control and Repository Documentation – Available at: <https://docs.github.com/>
- Flask/Python Web Development Reference Materials and Online Community Tutorials.
- Various online articles and technical blogs related to job recommendation systems and web application deployment.
- College lecture notes, practical assignments, and classroom guidance received during the B.C.A. course.
- Project source code repository maintained on GitHub for development and deployment support.